

Project 6

Kinematically-Redundant Planar Parallel Manipulator - 6DOF

Format: Group of two students; simulation-based project.

Motivation and context

Kinematic redundancy allows a robot to avoid singularities that would otherwise limit its workspace and degrade its accuracy. Indeed, the extra degrees of freedom can be used to select favorable configurations, avoid joint limits and obstacles, and maintain a well-conditioned Jacobian even near configurations that would be singular for a non-redundant robot.

A kinematically-redundant parallel robot is shown in Fig. 1. The robot is also shown in the online video <https://youtu.be/bJsRtkE2CVQ>. Three leg mechanisms are attached to an articulated platform by spherical joints. Each leg is able to control the position of its corresponding spherical joint. The articulated end-effector platform has three passive revolute joints with parallel axes. The robot is therefore kinematically redundant: the end-effector has three additional DOFs related to the revolute joints.

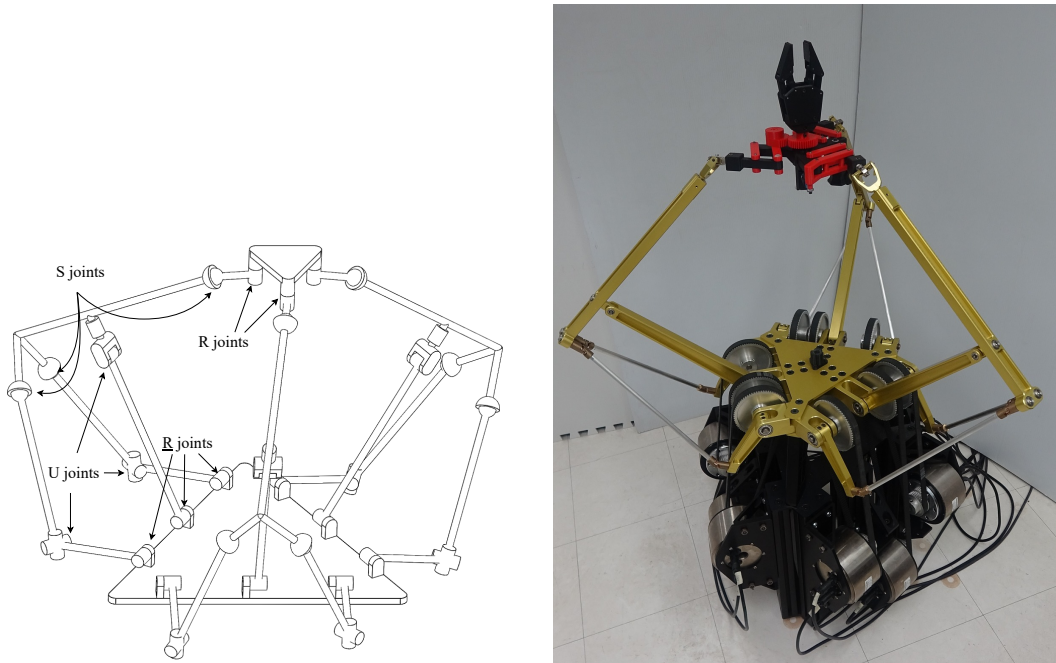


Figure 1: CAD model and prototype of the kinematically-redundant robot [1].

In this project, a planar case is considered. The leg mechanisms from [1] are replaced by 5-bar mechanisms and are connected to the articulated platform by revolute joints rather than spherical joints. The resulting architecture is shown in Fig. 2.

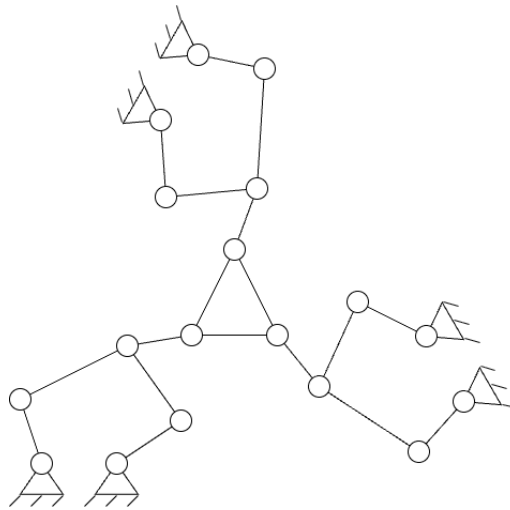


Figure 2: Kinematics architecture of the actuation-redundant robot.

Tasks

Overall objectives. Perform a singularity analysis of the entire robot. Show that one of the revolute joints on the platform can undergo unlimited rotation with appropriate redundancy resolution strategies.

- **Compute forward and inverse kinematics.** While inverse kinematics can be solved analytically, numerical methods can be used for forward kinematics.
- **Create a Simscape model of the kinematically-redundant robot.** Use this model in order to verify the kinematics equations.
- **Analyze the singularities.** You can use the results from [2], giving conditions for parallel singularities on the reconfigurable platform. However, note that singularities associated to the leg mechanism are not considered in this paper. Use results from Grassmann geometry to provide a geometrical interpretation of singularities.
- **Design a trajectory in augmented task space.** This trajectory should illustrate the unlimited rotation capability. Hint: divide the trajectory into constant-velocity sequences.
- **Determine joint torques.** Consider quasi-static assumption with mass associated to the end-effector and gravity. Are the simulation results consistent with the singularity analysis?

Available documents

Papers [1, 2] can be accessed online.

Expected outcome

Students should obtain the forward and inverse kinematics and the Jacobian matrices of the kinematically-redundant robot. They should understand why and how kinematic redundancy allows for doing unlimited rotation.

Submit a report, and a short video (screen recording) with Simscape illustrating unlimited rotation.

References

- [1] A. Yiğit, S. Foucault, T. Laliberté, D. Breton, Z. Zhou, and C. Gosselin, “Parallel robot with proprioceptive actuation for low-impedance sensorless pHRI,” in *Control Engineering Practice*, 2026, doi: 10.1016/j.conengprac.2026.106800.
- [2] A. Yiğit, D. Breton, and C. Gosselin, “Exploiting the kinematic redundancy of a (6+3)-DoF parallel manipulator to produce unlimited rotation of the platform,” in *ASME Journal of Mechanisms and Robotics*, 2024, doi: 10.1115/1.4063407.